Jacob Lontoc

Student at UMBC – B.S. Computer Engineering Major, Physics Minor
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Education

Computer Engineering B.S. & Physics
University of Maryland, Baltimore County
Aug 2024 – May 2028

Experience

Orivia Tech | Building Intelligence Engineer, Lead
- Four-month long fellowship for hardware and software engineering.
- Created hardware product that uses sensors to collect data from homes to create informed energy-related recommendations for homeowners, building managers, and energy auditors.

Vostner | Co-founder
- Private AI software, server orchestration, on-premises deployment
- Develop software for deploying and managing private AI on company-owned hardware. Design custom servers and support existing infrastructure, giving businesses data governance, models, and workloads.

Blue Wave Semiconductor | Electrical Engineering, Intern
- Internship focused on manufacturing R&D semiconductors fabricators for labs and universities.
- Gained experience in CNC and team-involved CAD design. Developed parts for a hot filament CVD and lab-grown diamond fabricator. Involved in electrical engineering and mentoring new interns.

McDonald's | Crew Trainer
- Trained new crew members on kitchen stations, food safety standards, and operating procedures.

Projects

ML Systems Research | Researcher
- Conducting collaborative research on MoE inference optimization, focusing on GPU memory management and decode latency.
- Developing a vLLM-based implementation of VAMP's dynamic memory repartioning framework.
- Utilizing predictive expert prefetching and compressed weight transfer to recover degraded decoding performance while preserving time-to-first-token improvements.

IEEE Sumobot Competition | Project Leader
- Placed first place against 15 other teams. Led the team in the programming phase
- Soldered components together, 3D-modeled chassis pieces, programmed robot logic ground up
- General understanding of infrared, Arduinos, and engineering design process

Skills

Programming C/C++, Python, Kotlin, Bash, JS
Embedded & Hardware Embedded C, Microcontrollers (ESP, STM, Arduino), HMI (Maple Systems, Modbus), Sensor Interfacing (I2C, SPI, UART, IMUs), Lasers, Soldering, Fusion 360
AI & Machine Learning PyTorch**,** vLLM/Tensor-RT-LLM, ONNX, FlashAttention, Model Quantization, Inference Optimization, Fine-tuning (LoRA)
Platforms Linux/Unix, Docker, AWS, Cloudflare, SOC2/HIPAA, MATLAB, Proxmox, Git